MIL-B-60674(MU)
24 January 1967

MILITARY SPECIFICATION
BOOSTER M21A4E1 ASSEMBLY
LESS BOOSTER PELLET AND LEAD CUP CHARGE

1. SCOPE

1.1 This specification covers one type of booster for use with high explosive ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-331	- Fuze and Fuze Components Environmental and Performance Tests for
MIL-STD-639	- Surface Roughness Values Preferred for Army Ordnance Materiel
MIL-STD-1167	- Ammunition Data Card

DRAWINGS

Munitions Command

F10520693	- Booster, M21A4E1 Assembly
C7548175	- Box, Packing Ammunition, for Booster Assembly, Details and Packing
IEL-10520693	- Inspection Equipment List for Booster, M21A4E1

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

FSC 1390

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3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on applicable drawings, referenced standards, and specifications except as provided in 6.1.

3.2 Environmental.

3.2.1 The booster shall comply with MIL-STD-331, Jolt, test No. 101.

3.2.2 The booster shall comply with MIL-STD-331, Jumble, test No. 102.

3.2.3 The booster shall comply with MIL-STD-331, Transportation-Vibration, test No. 104.

3.3 Non-arming.- The booster shall not arm when spun at the non-arming speed specified on the applicable drawing.

3.4 Arming.- The booster shall arm when spun at the arming speed specified on the applicable drawing.

3.5 Ballistics.- (see 6.3.2)

3.5.1 The booster shall not function prematurely. (see 6.3.1)

3.5.2 Target impact nonfunction distance.- The booster shall not detonate the round upon impact element function when fired in the 75mm Gun, Service Charge, (2800 feet per second, f.p.s.) on a target 180 plus 5 feet from the muzzle of the weapon.

3.5.3 Target impact function distance.- The booster shall detonate the round upon impact when fired in the 105mm Howitzer, charge 7 (1550 f.p.s.) on a target 400 minus 5 feet from the muzzle of the weapon.

3.5.4 Ground or water impact.- The booster shall detonate the round upon impact when fired in the 155mm Howitzer, charge 1 (680 f.p.s.) at an impact angle of 45° minimum.

3.6 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects.- All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.6.2 Surface finish.- The requirements for surface finish are as detailed on the applicable drawings and MIL-STD-639.

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3.6.3 Threads.- Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawing.

3.6.4 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.5 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter. The method and agents used in processing cleaning shall not adversely affect the finish or contaminate the parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.2 First article sample.

4.2.1 When required by the contract, the first units of production shall be submitted as an initial production sample. The sample shall consist of the following:

20 component parts as delineated in 4.2.2
75 subassemblies as delineated in 4.2.3
58 assemblies as delineated in 4.2.4 and 4.2.5

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection by the contractor and provisional acceptance at source by the Government, 45 booster assemblies shall be forwarded to a designated proving ground as specified in the contract. Twenty (20) component parts, 75 subassemblies and 13 booster assemblies shall be inspected as provided herein, at a Government laboratory or such other facility as specified in the contract.

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4.2.2 Component parts.

4.2.2.1 Twenty (20) components shall be tested as in 4.3.3.1 (Hardness).

4.2.3 Subassemblies

4.2.3.1 Twenty-five (25) of each subassembly shall be tested as in 4.3.3.2 (Static load).

4.2.3.1.1 Acceptance criteria.- Failure of any subassembly to comply with the requirements of the applicable drawing shall be cause for rejection of the initial production sample.

4.2.4 Environmental Tests.- Thirteen (13) booster subassemblies shall be subjected to the following MIL-STD tests:

Test	Quantity	Requirement	Test Method	Acceptance Criteria
Jolt	4	3.2.1	4.4.5.1	4.2.4.1.1
Jumble	4	3.2.2	4.4.5.2	4.2.4.1.1
Transportation-Vibration	5	3.2.3	4.4.5.3	4.2.4.1.2

4.2.4.1 Acceptance criteria.- Occurrence of any of the following in any booster shall be cause for rejection of the initial production sample.

4.2.4.1.1 Jolt and jumble

a. Rotor armed or partially armed. (see 6.2.1 and 6.2.2)

b. Broken, deformed or displaced parts which render the booster unsafe to handle or dangerous to use.

4.2.4.1.2 Transportation-vibration.

a. Condition listed in 4.2.4.1.1

b. Failure of the booster to meet the arming and non-arming requirement of the applicable drawing.

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4.2.5 Ballistics.- Forty-five (45) boosters, of the initial production sample shall be completely loaded and subjected to the following ballistic test phases.

Phases	Quantity	Requirement	Test Method
Impact Nonfunction Distance	15	3.5.2	4.4.6.1
Impact Function Distance	15	3.5.3	4.4.6.2
Ground or Water Impact	15	3.5.4	4.4.6.3

4.2.5.1 Acceptance criteria.- The occurrence of the following shall be cause for rejection of the initial production sample:

a. A premature burst in any phase (see 6.3.1)

b. Function of one or more rounds upon target impact in the Impact Nonfunction Phase

c. Failure of more than one round total to function in either of the following phases:

(1) Target Impact Function Phase

(2) Ground or Water Impact Phase

4.3 Inspection provisions.

4.3.1 Submission of product.- The product shall be presented in accordance with MIL-STD-105.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with MIL-STD-105. For the classification of defects contained herein, the AQL given for each shall be used. All nonconforming material shall be rejected. The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burrs and foreign matter.

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4.3.2.1 Cup, Booster, Drawing B8595509 covering a detail of Drawing F10520693.

Defect	Method of Inspection
CRITICAL	
None defined.	
MAJOR AQL = 1.0%	
101. Inside diameter, improper	Gage
102. Thickness of bottom, improper	Gage
103. Inside depth, improper	Gage
104. Curvature of bottom exceeds max.	Gage
105. Pitch diameter of thread, improper	Gage
106. Minor diameter of thread exceeds max.	Gage
107. Eccentricity of outside diameter with thread exceeds max.	Gage
108. Outside diameter, improper	Gage
109. Effective length of thread, less than min.	Gage
110. Metal defective	Visual
111. Cut, score, or gouge on inside wall	Visual
112. Foreign matter	Visual

4.3.2.2 Cup, Closing, Drawing B8595508, covering a detail of Drawing F10520693.

CRITICAL	
1. Split, cut or crack	Visual
MAJOR AQL = 1.0%	
101. Thickness of base, improper	Gage
102. Length under flange, improper	Gage
103. Outside diameter exceeds max.	Gage
104. Score, wrinkle or dent	Visual
105. Foreign matter	Visual

4.3.2.3 Body (Modified), Drawing D10520695 covering a detail of Drawing F10520693.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Eccentricity of large datum diameter with pitch diameter of internal thread	Gage
102. Eccentricity of small datum diameter with pitch diameter of internal thread	Gage

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4.3.2.3 MAJOR Cont'd

	Defect	Method of Inspection
103.	Eccentricity of diameter at bottom of cavity with pitch diameter of internal thread	Gage
104.	Pitch diameter of internal thread	Gage
105.	Minor diameter of internal thread	Gage
106.	Depth to small datum diameter	Gage
107.	Depth to large datum diameter	Gage
108.	Thickness through base	Gage
109.	Centrifugal pin screw missing or protrudes above flush	Visual-Gage
110.	Metal defective	Visual
111.	Foreign matter	Visual

4.3.2.4 Pallet Assembly, Drawing D10523174, covering a detail of Drawing F10520693.

CRITICAL

1.	Insecure or missing pallet pins	Manual and Visual

MAJOR AQL = 1.0%

101.	Height of pins less than min.	Gage
102.	Pallet inadequately staked or seated on pallet shaft	Visual
103.	Metal defective	Visual
104.	Foreign matter	Visual

4.3.2.5 Escapement Wheel and Pinion Assembly D10523170, covering a detail of Drawing F10520693.

CRITICAL

None defined

MAJOR AQL = 1.0%

101.	Gear inadequately staked or seated on pinion	Visual
102.	Gear tooth or pinion leaf missing	Visual
103.	Tooth, leaf or pivot deformed	Visual
104.	Metal defective	Visual
105.	Foreign matter	Visual

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4.3.2.6 No. 1 Gear and Pinion Assembly Drawing D10523167 covering a detail of Drawing F10520693.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Gear inadequately staked or seated on pinion	Visual
102. Gear tooth or pinion leaf missing	Visual
103. Tooth leaf pivot deformed	Visual
104. Metal defective	Visual
105. Foreign matter	Visual

4.3.2.7 Rotor assembly, Drawing D10523017, covering a detail of drawing F10520693.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Insecure, improper or incomplete assembly	Manual/Visual
102. Detonator installed incorrectly (with yellow end at crimp)	Visual
103. Detonator extends above flush of rotor surface	Visual
104. Rotor rivet extends above flush of rotor surface	Visual
105. Staking or crimping of detonator assembly inadequate	Visual
106. Metal burred, damaged or defective	Visual

4.3.2.8 Booster M21A4E1 Subassembly, Drawing F10520694, covering a detail of drawing F10520693.

CRITICAL	
1. Interference condition which precludes return of either spring-loaded block from the fully open position to the position of engagement in the rotor recess. (Check prior to nonarming test)	Visual and Manual
2. Rotor not locked in unarmed position (See 6.2.3) (Check immediately prior to packing.)	Visual and Manual

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4.3.2.8 Cont'd

Defect		Method of Inspection
MAJOR	AQL =.40%	
101.	Length from base of body to top plate exceeds max.	Gage
102.	Protrusion above the movement plate of all screw and pins, exceeds max.	Gage
103.	Lock pin screw extends above flush of bottom surface of body	Gage
104.	Insecure, improper or incomplete assembly	Visual and Manual
105.	Staking missing or inadequate	Visual
106.	Burr or bulge on bottom surface of body	Visual
107.	Metal defective	Visual
108.	Threads damaged or not full	Visual
109.	Foreign matter	Visual

4.3.2.9 Carton, packing, drawing C7548174.

CRITICAL

None defined

MAJOR	AQL = 1.0%	
101.	Assembly damaged so that contents are exposed or liable to become exposed	Visual
102.	Sealing strip incomplete, badly wrinkled, or not in contact with surface of containers for at least 1/4 inch around entire circumference.	Visual
MINOR	AQL = 2.5%	
201.	Contents move when shaken.	Manual
202.	Marking misleading, incorrect or illegible	Visual
203.	Label missing	Visual

4.3.2.10 Box, packing, drawing C7548175.

CRITICAL

None defined

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4.3.2.10 Cont'd

Defect		Method of Inspection
MAJOR	AQL = 1.0%	
101.	Board broken	Visual
102.	Hardware missing, broken or loose	Visual/Manual
103.	Contents exposed	Visual
104.	FSC symbol misleading, incorrect or illegible	Visual
105.	Strapping missing, broken or loose	Visual/Manual
MINOR	AQL = 2.5%	
201.	Hardware improperly applied or engaged	Visual/Manual
202.	Contents move when shaken	Manual
203.	Car seal missing, unsealed or improperly positioned.	Visual/Manual
204.	Marking misleading, incorrect or illegible	Visual/Manual

4.3.2.11 Process control audit.- Periodic unscheduled monitoring shall be conducted for the process operations on the following assemblies to determine compliance with the requirements of the applicable drawings.

a. Movement assembly, drawing F10523008.

(1) Application of lubricant
(2) Staking operation on assembly screws

b. Booster, M21A4E1 Subassembly, Drawing F10520694

(1) Staking operation on assembly screws.

4.3.3 Inspection testing.- The following test shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Hardness (destructive).- Twenty pallet pins from each lot shall be subjected to the hardness specified on drawing B10523175. Failure of any pin to comply with the requirements shall be cause for rejection of the lot of pallet pins (See 4.4.1).

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4.3.3.2 Static load (destructive).- Twenty-five samples from each lot of the following subassemblies shall be subjected to the static load test. Failure of any subassembly to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot of subassemblies, as applicable (See 4.4.2).

Pallet Assembly (Pins and shaft)	D10523174
Escapement Wheel and Pinion Assembly	D10523170
No. 1 Gear and Pinion Assembly	D10523167

4.3.3.3 Arming (nondestructive).- Each booster assembly shall be tested to determine compliance with the arming requirement of drawing F10520694. Failure to comply with the arming requirement shall be cause for rejection of the assembly (See 4.4.3).

4.3.3.4 Nonarming (nondestructive).- Each booster assembly shall be tested to determine compliance with the nonarming requirement of drawing F10520694. Failure to comply with the requirement shall be cause for rejection of the assembly (See 4.4.4).

4.3.3.5 Environmental tests (destructive).- Thirteen (13) booster assemblies from each production lot shall be subjected to the following MIL-STD tests (See 4.4.5).

Test	Quantity	Requirement	Test Method	Acceptance Criteria
Jolt	4	3.2.1	4.4.5.1	4.3.3.5.1.1
Jumble	4	3.2.2	4.4.5.2	4.3.3.5.1.1
Transportation-Vibration	5	3.2.3	4.4.5.3	4.3.3.5.1.2

4.3.3.5.1 Acceptance criteria.- Occurrence of any of the following in any booster shall be cause for rejection of the production lot.

4.3.3.5.1.1 Jolt and Jumble.-

a. Rotor armed or partially armed (See 6.2.1 and 6.2.2)

b. Broken, deformed or displaced parts which render the booster unsafe to handle or dangerous to use

4.3.3.5.1.2 Transportation-Vibration.-

a. Conditions listed in 4.3.3.5.1.1

b. Failure of the booster to meet the arming and nonarming requirement of the applicable drawing.

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4.3.3.6 Ballistics (destructive).- Forty-five (45) boosters from each lot shall be completely loaded and subjected to the following ballistic test phases. (See 4.4.6).

Phases Ambient	Quantity	Requirement	Test Method
Impact Nonfunction Distance	15	3.5.2	4.4.6.1
Impact Function Distance	15	3.5.3	4.4.6.2
Ground or Water Impact	15	3.5.4	4.4.6.3

4.3.3.6.1 Acceptance criteria.- The occurrence of the following shall be cause for rejection of the lot the ballistic sample represents:

a. A premature burst in any phase (See 6.3.1)

b. Function of one or more rounds upon target impact in the Impact Nonfunction Phase.

c. Failure of more than one round total to function in the combined following phases:

(1) Target Impact Function Phase

(2) Ground or Water Impact Phase

4.3.4 Inspection equipment.- The examinations and tests shall be made using equipment listed in IEL-10520693. Equipment design shall be in accordance with the code designation on the applicable IEL.

4.4 Test method and procedures.

4.4.1 Hardness (Pin, Pallet).- The test shall be performed using the value as specified on drawing B10523175. Samples shall be tested in an area which is solidly supported in order to obtain valid readings. The median value of three hardness readings taken on each sample unit shall be used as the individual reading. All tested samples shall be scrapped.

4.4.2 Static load.- All units shall be tested to the value specified on the applicable drawing. All units subjected to the static load test shall be scrapped.

4.4.3 Arming.- The unarmed booster shall be rotated for a period of 10 seconds maximum in a counterclockwise direction when viewed from the top, at a constant r.p.m. as specified on drawing F10520694. At the conclusion of the test, the rotor shall be returned to the unarmed position (See 6.2.1 and 6.2.3).

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4.4.4 Nonarming.- The unarmed booster (see 6.2.3) shall be rotated in a counterclockwise direction when viewed from the top, at a constant r.p.m. as specified on drawing F10520694. The nonarming test shall be the last spin test to which the booster is subjected prior to packing.

4.4.5 Environmental Test Procedures.- The component parts of the booster, which have been disassembled and inspected following tests, shall be scrapped. Where spin testing is required, the nonarming test shall be performed prior to the arming test (drawing F10520694).

4.4.5.1 Jolt.- The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 101. The booster shall be disassembled and inspected as follows:

a. Remove dust cover check position of rotor.

b. Check both safety blocks (by probing the blocks for the fully open position to engagement in the rotor recess).

c. Remove top plate.

d. Check for broken, deformed or displaced parts (that would render the booster unsafe to handle a dangerous to use).

4.4.5.2 Jumble.- The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 102. The booster shall be disassembled and inspected as follows:

a. Conditions listed 4.4.5.1

4.4.5.3 Transportation-Vibration.- The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 104.

a. Remove dust cover check position of rotor.

b. Subject the booster to the nonarming and arming spin in that order.

c. Conditions listed in 4.4.5.1 (c and d)

4.4.6 Ballistics.- The booster assembly, drawing F10520693, shall be assembled to Fuze, PD, M48A3 then assembled to a suitable projectile and fired as follows:

4.4.6.1 Impact nonfunction distance phase.- The test sample shall be fired as specified in 3.5.2 against a suitable supported one inch plywood screen of adequate size under the conditions delineated in 4.2.5 or 4.3.3.6 as appropriate. Observation of impact element function is required.

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4.4.6.2 Impact function distance phase.- The test sample shall be fired as specified in 3.5.3 against a suitably supported one inch plywood screen of adequate size under the conditions delineated in 4.2.5 or 4.3.3.6 as appropriate.

4.4.6.3 Ground or water impact phase.- The test sample shall be fired as specified in 3.5.4 under the conditions as delineated in 4.2.5 or 4.3.3.6 as appropriate.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, labeling and marking.- Packaging, packing, labeling and marking shall be as specified on Drawing C7548175 with the following exception. Closing Cups, Drawing B8595508 and Booster Cups, Drawing B8595509 may be shipped separately in accordance with best commercial practices as an alternate procedure.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Components manufactured in strict accordance with applicable drawings represent one set of conditions which will result in satisfactory performance. Dimensions which are specifically required by Classification of Defects, as listed in Section 4 and those which affect external profile of assembly, are mandatory. Pertaining to other dimensions, it is recognized that current mass production methods may occasionally produce parts which are not in full compliance with all dimensional requirements, but which may provide adequate booster performance. Therefore, the contractor, at his own risk, may utilize these parts provided there is no adverse effect on booster functionability. This relaxation of requirements applies to "after the fact" deviations only, to dimensions only, on applicable drawings, and does not constitute authority to establish new dimensions or to relax efforts to achieve the specified limits. The procedure for accomplishing dimensional changes is available under the established engineering change system. Nothing above is to be construed as relieving the contractor of any responsbility in connection with performance of the booster.

6.1.1 The contractor shall maintain records showing deviations from dimensional requirements, the extent of such deviations, the amount of material involved, and the date the material is introduced into the assembly line. Copies of these records shall be forwarded to the agency assigned custody of the technical data.

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6.2 Armed/Unarmed Boosters.

6.2.1 Armed.- Rotor is locked by the spring loaded detent pin in the position in which the detonator is fully visible through the flash hole.

6.2.2 Partially armed.- Rotor not locked by both spring-loaded safety blocks.

6.2.3 Unarmed.- Rotor locked by both spring loaded safety blocks with the detonator not at all visible through the flash hole.

6.3 Ballistic Parameters.

6.3.1 Premature Burst.- Shall be defined as the occurrence of any of the following:

a. If the round detonates in the bore of the weapon.
b. If the round detonates in flight within 180 feet from the muzzle of the weapon.

6.3.2 Muzzle velocities mentioned in this specification are nominal only and are given for the guidance of the proving ground in selecting appropriate weapons and rounds.

6.4 Ordering data.- Procurement documents should specify the title, number and date of this specification.

Custodian:

Army-MU

Preparing activity:

Army-MU

Project No. 1390-A-603

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